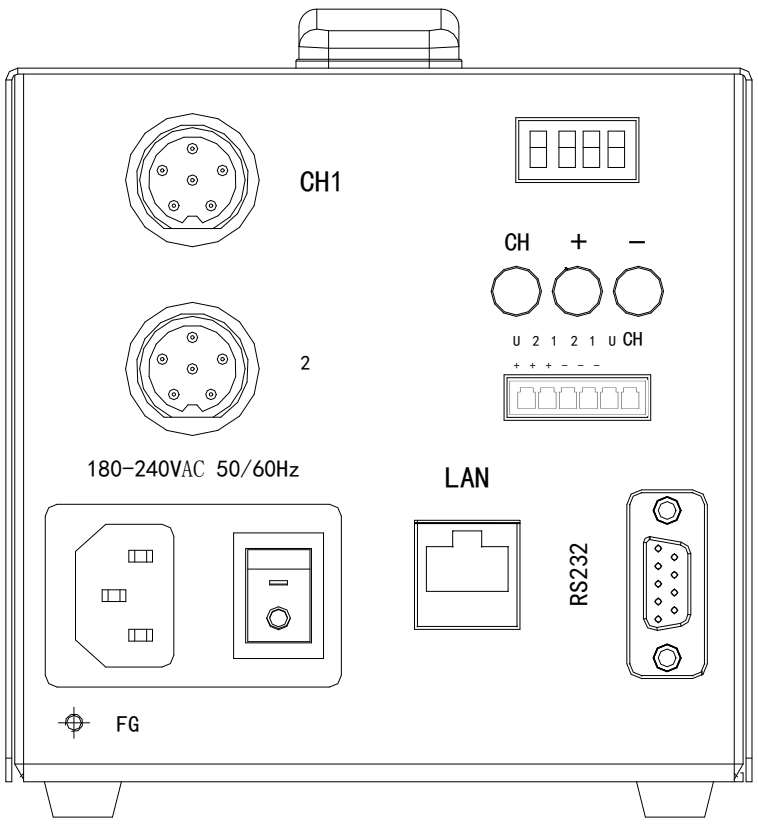
东莞锐视光电科技有限公司
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Dongguan Rsee Lighting Technology Co., Ltd.Address: No.333 Zhushan Zhenxing Road, Dongcheng District, Dongguan City
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P-AHC-48W600-2T 大功率恒流控制器使用说明书_V2022
P-AHC-48W600-2T High-Power Constant Current Controller User Manual_V2022



■ 产品特点
■ Product Features

● 提供手动/RJ45 网口/RS232 串口通讯方式控制光源。

● Supports manual, RJ45 network port, or RS232 serial communication methods for controlling the light source.

● 提供外部 IO 触发端口，按需点亮光源，延长光源寿命。

● Provides external I/O trigger ports for activating the light source on demand, thereby extending its lifespan.

● 具有断电记忆、短路保护、过载保护等功能。

● Features include power-off memory, short-circuit protection, and overload protection.

● 采用恒流驱动方式，使光源发光更稳定。

● The constant-current drive method ensures more stable light emission from the light source.

● 自动识别光源电流，兼容同系列不同规格光源。

● Automatically identifies the light source current and is compatible with light sources of different specifications within the same series.

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安全须知
Safety Instruction

电气安全须知
Electrical Safety Guidelines

- ✧ 本光源控制器只能用于驱动恒流型 LED 负载，不可用于驱动其它负载，若违规使用造成的损失由客户自己承担。
- ✧ This light source controller is only intended for driving constant-current LED loads and shall not be used to drive other types of loads; any losses resulting from improper use shall be borne solely by the customer.

- ✧ 为避免可能的电击造成严重损害，在搬动控制器之前，请先将电源线从电源插槽中拔掉。
- ✧ To prevent potential electric shock that could cause serious injury, please unplug the power cord from the power slot before moving the controller.

- ✧ 当您要连接其它硬件设备到控制器时，请先切断控制器输入电源，通电前请仔细检查输入、输出接线是否连接正确，方可确保控制器及外围设备可靠工作。
- ✧ When you need to connect other hardware devices to the controller, first disconnect the controller's input power supply; before powering on, carefully verify that all input and output wiring is correctly connected to ensure the proper and reliable operation of both the controller and its peripheral devices.

- ✧ 请您确定电源供应的电压设置已经调整到本国或本区域所使用的电压标准值。若不确定您所属区域的供应电压值，那么请就近询问当地的电力公司人员。
- ✧ Please ensure that the power supply voltage setting has been adjusted to the standard voltage value used in your country or region. If you are unsure of the supply voltage for your area, please consult a representative from your local power company.

- ✧ 为确保使用安全性及提高抗干扰性，请确保输入电源的地线可靠接地。
- ✧ To ensure operational safety and enhance anti-interference performance, please ensure that the ground wire of the input power supply is reliably grounded.

操作安全须知
Operating Safety Instructions

- ✧ 在使用该产品前，请仔细阅读该使用说明书。
- ✧ Before using this product, please carefully read the instruction manual.

- ✧ 在使用该产品前，请您确定产品的外观等质量，如发现任何重大的瑕疵，请尽快联络您的经销商。
- ✧ Before using this product, please verify its appearance and other quality aspects; if any major defects are identified, please contact your distributor as soon as possible.

- ✧ 灰尘、湿气以及剧烈的湿度变化都会影响控制器的使用寿命，因此请尽量避免放置在这些地方。
- ✧ Dust, moisture, and significant fluctuations in humidity can all affect the service life of the controller; therefore, please avoid placing it in such environments.

- ✧ 请勿将控制器放置在容易摇晃的地方。
- ✧ Do not place the controller in a location prone to shaking.

- ✧ 若在本使用说明书上有任何的技术性问题，请和经过检定或有经验的技术人员联系。
- ✧ If you have any technical questions regarding this manual, please contact a certified or experienced technician.

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■ 产品规格

■ size of product

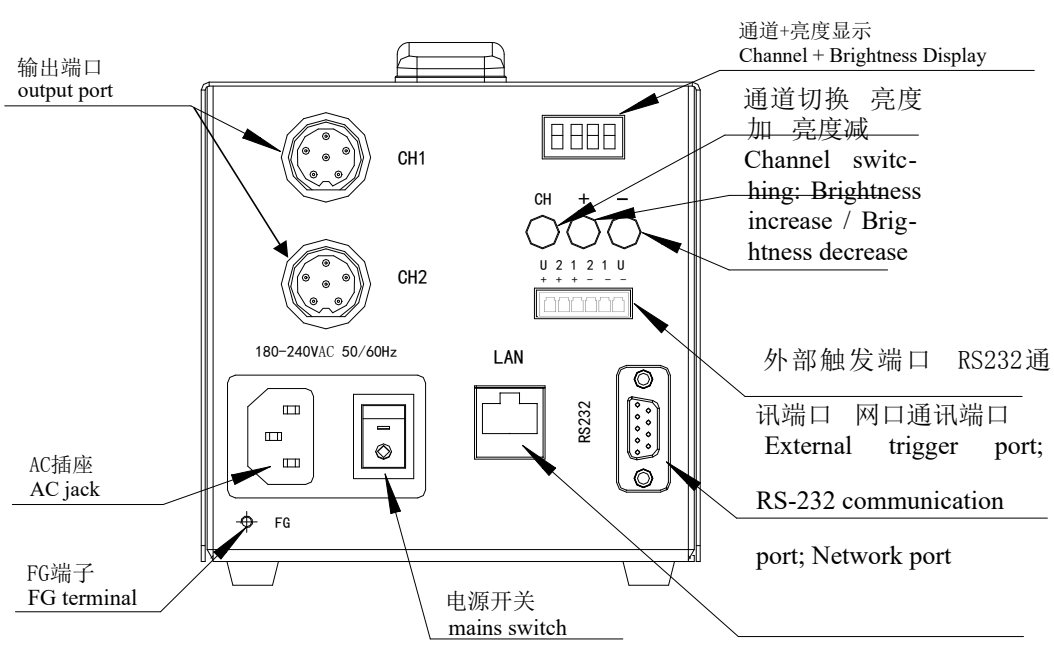
产品型号 Product model		P-AHC-48W600-2T
驱动方式 Drive method		恒流驱动 Constant current drive
调光方式 Dimming method	手动 Hand movement	面板按键 256 级调光 Panel buttons: 256-level dimming
	外部 Outside	RJ45 网口通讯/RS232 串口通讯 RJ45 Network Interface / RS232 Serial Communication
输入电压 Input voltage		180-240VAC 50/60Hz 680VA
输出通道 Outgoing channel		2
输出参数 Out parameter		DC48V 总电流 12A(单通道 6A) DC48V total current: 12A (single channel: 6A)
适用光源(额定) Applicable light source (rated)		≤600W
触发延迟 Trigger delay		≤150us
外触发电压 External trigger voltage		DC5-24V（最大电流约 5.6mA） DC 5–24 V (maximum current: approximately 5.6 mA)
输出端口 Output port		19M-6G
使用温湿度 Use temperature and humidity controls		温度：0-+40℃、 湿度：20-85%RH（非凝结） Temperature: 0–+40 °C; Humidity: 20–85% RH (non-condensing)
保存温湿度 Preserve temperature and humidity		温度：-20-+60℃、湿度：20-85%RH（非凝结） Temperature: -20–+60 °C; Humidity: 20–85% RH (non-condensing)
冷却方式 Cooling-down method		强制冷却(内置风扇) Forced cooling (built-in fan)
环境管制 Environmental Regulation		RoHS

■ 使用方法

■ usage method

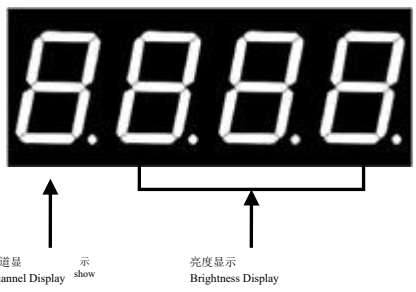
●前面板

●front panel



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●显示与按键说明
●Display and Button Instructions



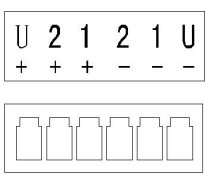
通道显示 Channel Display	
显示值 Show value	说明 Explain
1	CH1 通道 CH1 channel
2	CH2 通道 CH2 channel
操作按键：CH 按键 Control button: CH Button	

亮度显示 Brightness display	
显示值 Show value	说明 Explain
XXX	当前通道亮度值 Current channel brightness value
H	常亮触发模式 Constant-on trigger mode
L	常灭触发模式 Constant-off trigger mode
操作按键：+/−按键 Control buttons: +/- buttons	

备注：常亮触发模式下，无触发信号时光源点亮，有触发信号时光源熄灭；常灭触发模式下，无触发信号时光源熄灭，有触发信号时光源点亮，且点亮时间跟随外部触发信号高电平的持续时间。
Note: In the constant-on trigger mode, when no trigger signal is present, the light source remains on; when a trigger signal is present, the light source turns off. In the constant-off trigger mode, when no trigger signal is present, the light source is turned off; when a trigger signal is present, the light source turns on, and its on-time corresponds to the duration of the high-level external trigger signal.

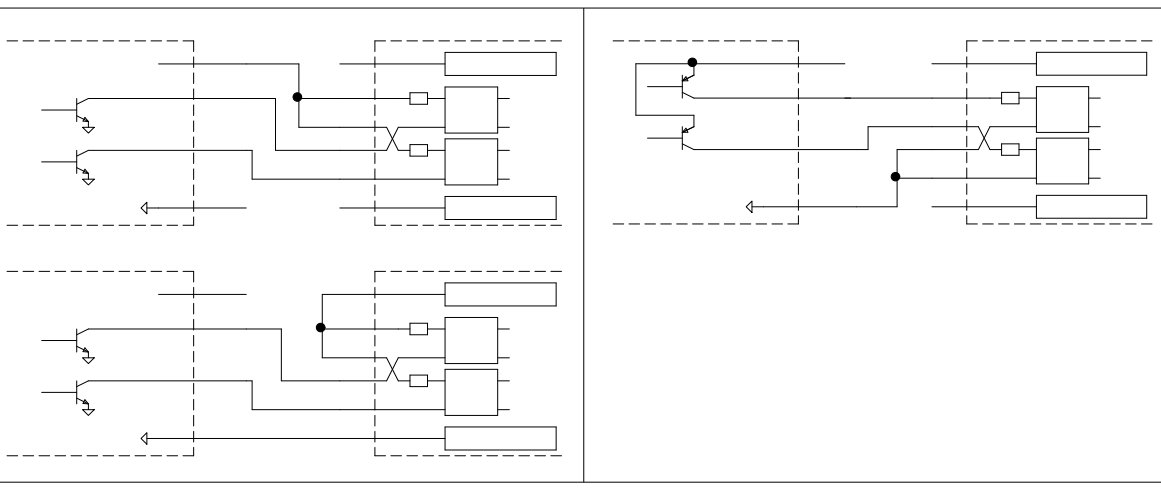
●外部触发输入
●External trigger input

外触发端口定义
External Trigger Port Definition



端口 Port	端口定义 Port Definition
U+	24V 电源输出正极 24V power supply output positive terminal
2+	CH2 外部触发输入正极 CH2 External Trigger Input Positive Pole
1+	CH1 外部触发输入正极 CH1 External Trigger Input Positive Pole
2−	CH2 外部触发输入负极 CH2 External Trigger Input Negative Terminal
1−	CH1 外部触发输入负极 CH1 External Trigger Input Negative Terminal
U−	24V 电源输出负极 24V power output negative terminal

外部触发接线简图
External Trigger Wiring Schematic



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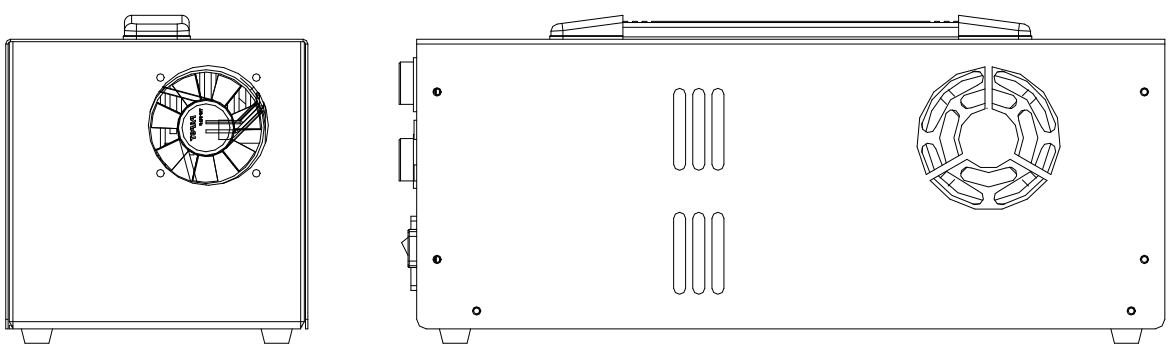
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●后面板
●Back Panel

控制器侧面和后面有散热进出风口，请勿堵塞风道，进出风口到旁边障碍物的距离必须确保 50mm 的空间间隙，便于控制器有效散热。

The controller's side and rear are equipped with heat dissipation air inlets and outlets; do not block these air passages. The distance between the air inlets/outlets and any adjacent obstacles must ensure a minimum clearance of 50 mm to facilitate effective heat dissipation for the controller.



●通讯协议说明
●Communication Protocol Description

本控制器支持外部 RS232、RJ45 局域网方式进行远程通讯控制。

This controller supports remote communication control via external RS232 or RJ45 LAN interfaces.

- 网口默认参数如下：
- The default parameters for the network port are as follows:

默认 IP 地址 Default IP Address	192. 168. 1. 252	网关地址 Gateway Address	192. 168. 1. 1
默认工作方式 Default operating mode	TCP server	默认端口号 Default port number	8234

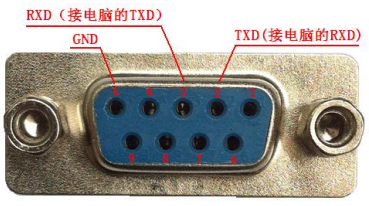
备注：电脑端工作方式设置成 TCP Client。

Note: Set the computer-based operating mode to TCP Client.

- 串口默认参数如下：
- The default parameters for the serial port are as follows:

在工作状态下，连接控制器上的 RS232 端口(从机类型)，按下面的协议标准，即可过发送指令的方式远程设置控制器各个参数，串口通讯协议如下：

During operation, connect to the RS232 port on the controller (slave mode), and follow the protocol standards below to remotely configure various parameters of the controller via command transmission; the serial communication protocol is as follows:

协议类型 Protocol type	RS232	串口引脚定义 Serial Port Pin Definition	
波特率 Baud rate	19200		
数据位 Data bit	8		
停止位 Stop bit	1		
校验位 Check bit	无 Not have		

- 指令说明(网口、串口指令一样)
- Instruction Description (Network port and serial port instructions are identical)

亮度设置 Brightness Setting	S	X	XXXX	#	通讯成功返回值 Successful communication return value
指令说明 Instruction Description	起始符 Start symbol	A:CH1 通道 A: CH1 Channel B:CH2 通道 B: CH₂ channel	0000-0255 (亮度等级) 0000–0255 (Brightness level)	结束符 Terminator	A(CH1 通道) A(Channel CH1) B(CH2 通道) B(CH2 channel)
亮度读取 Brightness Reading	S	X	#		xXXXX
指令说明 Instruction Description	起始符 Start symbol	A:CH1 通道 A: CH1 Channel B:CH2 通道 B: CH₂ channel	结束符 Terminator		x=a:CH1 通道 x=a: CH1 channel x=b:CH2 通道 XXXX:亮度值 x=b: CH2 channel XXXX: brightness value
触发模式设置 Trigger Mode Settings	S	X	#		H/L
指令说明 Instruction Description	起始符 Start symbol	H:常亮触发模式 H: Always-on trigger mode L:常灭触发模式 L: Constant-off trigger mode	结束符 Terminator		H:常亮模式 H: Always On Mode L:常灭模式 L: Constant Off Mode

注意事项：

Matters Need Attention：

1、每通道亮度值命令必须为四位十进制数字,通讯失败(指令无效、连接不成功等原因)无返回值；

1、 The brightness value command for each channel must be a four-digit decimal number;in the event of a communication failure (e.g., invalid command or unsuccessful connection), no return value will be provided;

2、例:发送SA0168#:设置CH1通道亮度值是168；发送SA0168# SB0168#:设置CH1、CH2的通道亮度值是168。发送SA#:读取CH1通道亮度值，发送成功后将收到返回值：a0100；

2、 Example: Send SA0168#: Sets the brightness value of CH1 to 168; Send SA0168# SB0168#: Sets the brightness values for both CH1 and CH2 to 168; Send SA#: Reads the brightness value of CH1; upon successful transmission, the system will return the following response value: a0100;

发送SH#:设置控制器为常亮触发模式；发送SL#:设置控制器为常灭触发模式。

Send SH#: Sets the controller to always-on trigger mode; Send SL#: Sets the controller to always-off trigger mode.

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网口设置如下：

The Network Port Settings Are as Follows:

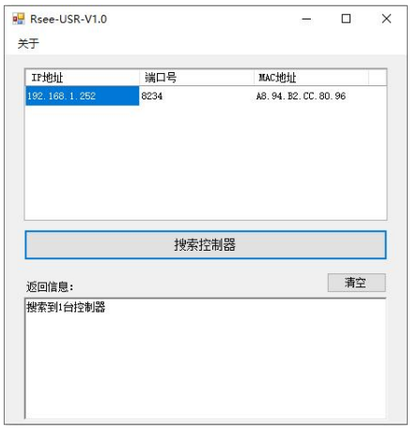
用户可通过我司提供的‘Rsee 网口配置工具(烧录 MAC 地址)’网口助手查询和设置控制器的网口参数。软件界面如下：

Users can use our provided 'Rsee Network Port Configuration Tool (MAC Address Programming)' network port utility to query and configure the controller's network port parameters. The software interface is shown below:



用网线连接控制器上的 RJ45 接口后，点击软件中的‘搜索控制器’按键，可搜索出已连接电脑上的控制器，如下：

After connecting the controller's RJ45 port to a network cable, click the' Search Controller' button in the software to search for controllers connected to your computer; the results will appear as follows:



双击 IP 地址栏下出现的 IP 地址，可弹出此 IP 端口的控制器网口参数，用户修改参数后点击右下方的‘保存设置’按键即可生效。如下：

Double-click the IP address displayed in the IP address field to open the controller's network port parameters for this IP port; after modifying the parameters, click the' Save Settings' button in the bottom-right corner to apply the changes. As shown below:



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■外形尺寸
■Outline Dimension

单位：mm
unit： mm

